(b) Explain what would happen if the human body touches

(i) the live wire of the mains supply in the primary circuit; (2 marks)

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(ii) one of the conducting wires in the shaver circuit outlet. (2 marks)

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*(c) What is the turns ratio of the primary coil to the secondary coil of the transformer so as to provide 110 V ? (1 mark)

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Answers written in the margins will not be marked.

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(b) Now a 7000 N counterweight is installed at the other end of the cable as shown in Figure 3.2. The counterweight always moves in the opposite direction to the lift car which again moves up at 2 m s^{-1}. Assume that there is no slipping between the cable and the drum.

(i) Calculate the total mechanical power output of the motor required in this case, assuming the same power loss in overcoming friction between movable parts as found in (a). (2 marks)

(ii) State the advantage of having the counterweight installed. (1 mark)

(iii) A claim is made that as power is lost due to friction, a drum with frictionless surface can further reduce the power required from the motor. Comment on this claim. (2 marks)

(b) Briefly explain the variation of the resistance of the light bulb with the voltage across the light bulb. (2 marks)

(c) The student claims that since the resistance of the light bulb is not a constant, the equation $R = V/I$ cannot be used to calculate the resistance of the light bulb. Briefly explain why his claim is wrong. (1 mark)

(d) Determine the resistance of the light bulb at $V = 0.1$ V and 2.5 V. (3 marks)

(e) It is given that the cross-sectional area of the tungsten filament in the light bulb is 1.66×10^{-9} m^2, and the resistivity of tungsten at room temperature is about 5.6×10^{-8} Ω m. Estimate the length of the tungsten filament in the light bulb using the appropriate resistance found in (d). (3 marks)

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Answers written in the margins will not be marked.

9. (a) Two long straight current carrying wires, P and Q, are parallel to each other and lie on the plane of the paper as shown in Figure 9.1. The currents in the wires, I_P and I_Q, flow in the same direction.

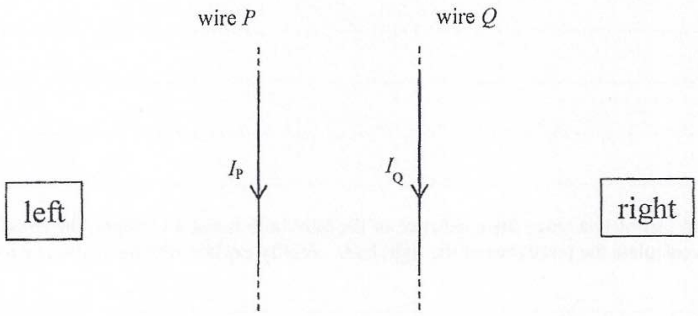

Figure 9.1

(i) State the direction (to the left / to the right / into the paper / out of the paper) of the magnetic field at Q due to P. (1 mark)

....

....

(ii) In Figure 9.1, draw the direction of the magnetic force acting on Q due to P. (1 mark)

(iii) Show that the magnitude of the magnetic force per unit length F_l acting on Q due to P is

$$F_l = \frac{\mu_o I_P I_Q}{2\pi r},$$

where μ_o is the permeability of free space and r is the separation between the two wires. (3 marks)

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(iv) For the magnetic force acting on Q due to P and the magnetic force acting on P due to Q, if $I_P \neq I_Q$, briefly explain whether the two forces are equal in magnitude. (2 marks)

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(c) The kitchen unit includes the following electrical appliances:

	rating	effective time of operation at rated value per day
a refrigerator	220 V, 500 W	8 hours
an electric kettle	220 V, 2000 W	0.5 hour
an induction cooker	220 V, 3000 W	2 hours

How much should be paid per day to run these appliances if 1 kW h of electrical energy costs \$0.9 ?
(3 marks)

Answers written in the margins will not be marked.

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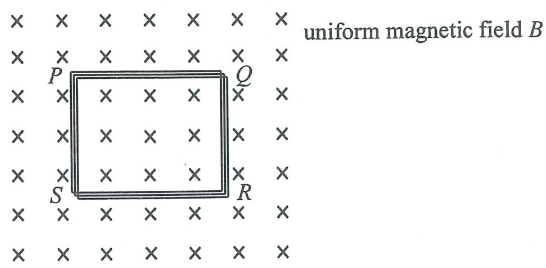
Answers written in the margins will not be marked.

9. A rectangular coil *PQRS* of 20 turns, each having an area of 0.005 m^2, is placed in a uniform magnetic field B of strength 0.3 T pointing into the paper as shown in Figure 9.1.

Figure 9.1



(a) The strength of the magnetic field decreases uniformly to zero within 0.5 s.

(i) Explain why a current would be induced in the coil. (2 marks)

*(ii) Calculate the change in total magnetic flux linkage through the coil and the value of the induced e.m.f. ξ in the coil. (3 marks)

Answers written in the margins will not be marked.

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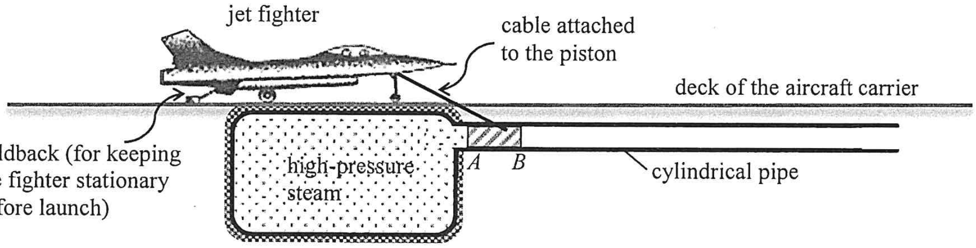
(b) This set-up can be used as a 'steam catapult' to launch jet fighters from an aircraft carrier. A jet fighter in position to be launched is connected to the piston via an inextensible cable as shown in Figure 2.2. When the holdback behind the jet fighter is released, the high-pressure steam in the gas tank expands and pushes the piston which in turn helps to accelerate the jet fighter.

Figure 2.2



In a trial run of the catapult, a jet fighter (with its engine shut down) acquires a final speed of 54 m s^{-1} in 1.5 s after running a distance horizontally on the deck. The mass of the jet fighter is 2.6×10^4 kg.

(i) Find the work done by the net force on the jet fighter during launch. (2 marks)

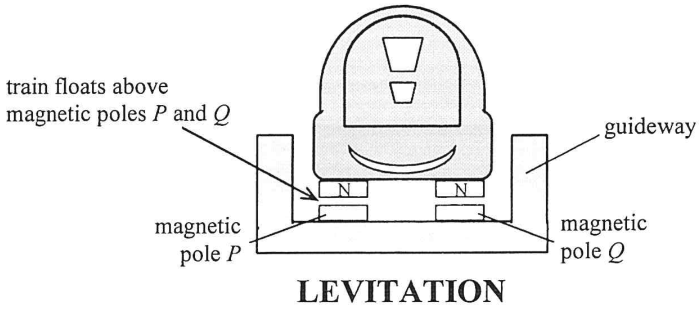
(ii) Calculate the average acceleration of the jet fighter during launch. (2 marks)

*(iii) State whether the acceleration of the jet fighter is increasing, decreasing or uniform during launch. Explain your answer. (3 marks)

Answers written in the margins will not be marked.

3. Read the following passage about a **magnetically levitated (maglev) train** and answer the questions that follow.

'A maglev train car is just a box with magnets on the four corners,' says Jesse Powell, the son of the maglev train inventor. The electromagnets employed have superconducting coils (i.e. coils with extremely low resistance). They therefore can generate magnetic fields 10 times stronger than ordinary electromagnets, enough to levitate and propel a train.



Two sets of magnetic fields are set up for different functions. One is to make the train float a few centimetres above magnetic poles P and Q as shown while the other is a propulsion system run by an alternating current for moving the train car along the guideway by magnetic attraction and repulsion. This floating design enables a smooth movement of the train. Even when the train travels up to 600 km per hour, passengers inside experience less vibration than travelling on traditional trains.

(a) Explain why electromagnets employing superconducting coils can produce much stronger magnetic fields. (2 marks)

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(ii) The amplitude of the CRO trace at P is **not** zero. Suggest a possible reason. (1 mark)

(c) Given: $AQ = 2.17$ m, $BQ = 2.58$ m
Find the speed of sound in air if the frequency of the signal generator is 1200 Hz. (2 marks)

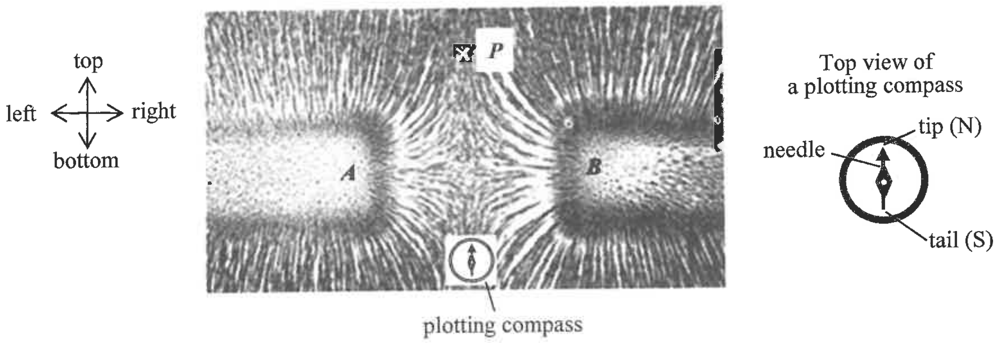
(d) Given that the separation of A and B is 0.80 m, explain why no more maximum can be detected when the microphone is moved beyond position R along OY. (2 marks)

(e) The microphone is now moved from O to X along the line OX. State whether the amplitude of the CRO trace increases, decreases, remains the same, or varies periodically. (1 mark)

7. Read the following passage about **magnetic field patterns** and answer the questions that follow.

Iron filings are tiny pieces of iron nearly in powder form. Since iron is ferromagnetic, a magnetic field would induce each piece of iron filings to become a 'tiny bar magnet'. The south pole of each of these 'tiny bar magnets' attracts the north poles of neighbouring iron filings. The magnetic field pattern is displayed as iron filings align themselves with the field lines.

The figure below shows such a pattern displayed on a cardboard with two identical bar magnets placed underneath. A plotting compass placed at the bottom of the figure points to the top as shown.



(a) (i) State the polarity of the respective poles at A and B of the bar magnets. (1 mark)

A : ....

B : ....

(ii) If the compass is moved to P, towards what direction (top, bottom, left or right) would it be pointing ? (1 mark)

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(iii) In obtaining such field patterns experimentally, one is advised to place the magnets **underneath the cardboard**. Why ? (1 mark)

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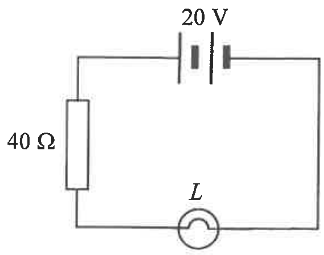
(ii) Explain why the resistance of the bulb varies with the applied voltage V. (2 marks)

(c) The bulb L in (b) and a 40 Ω resistor are connected in series with a 20 V battery of negligible internal resistance as shown in Figure 8.3.

Figure 8.3



The current I in the circuit and the voltage V across the bulb is related by $I = 0.5 - 0.025V$.

(i) Add a straight line on Figure 8.2 to determine the current I. (2 marks)

(ii) Hence estimate the power consumed by bulb L. (2 marks)

(iii) With S connected to terminal **2**, is the potential difference across R_2 greater than, smaller than or equal to that across R_1 ? (1 mark)

(iv) With S connected to terminal **1**, what is the power dissipated by the coil ? Give your reason. (2 marks)

Answers written in the margins will not be marked.

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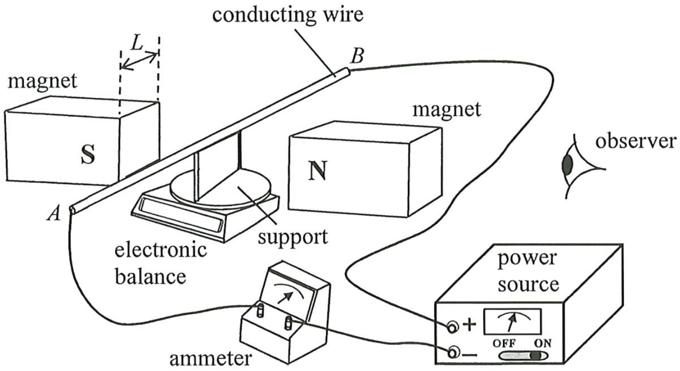
8. A student conducts an experiment to study the force on a straight current-carrying wire placed between the poles of a magnet providing a uniform magnetic field perpendicular to the wire. An adjustable power source is connected to the wire which is held horizontally by an insulated support placed on an electronic balance. The variation of the magnetic force on the wire is registered by the balance.

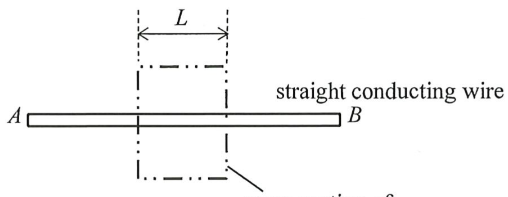
Figure 8.1



(a) On Figure 8.1, mark the positive (+) and negative (–) terminals of the ammeter. (1 mark)

(b) Figure 8.2 shows the side view of the wire AB seen by the observer. On the figure, draw the magnetic field provided by the magnet and indicate the directions of the current I and magnetic force F_B on the wire. (3 marks)

Figure 8.2



(c) Complete the table below to indicate how each of the factors listed would affect the magnitude of the magnetic force F_B (increase, decrease or unchanged). (3 marks)

	magnetic force F_B
using a stronger magnet of the same dimensions	
using another support such that the position of the horizontal wire is slightly lowered	
using a longer wire while the current is kept unchanged	

Answers written in the margins will not be marked.

(b) When the jockey is placed at point A (i.e. $l = 0$), find the resistance of the whole circuit. Compare it with the value of the 4.7 Ω resistor and account for the difference. (3 marks)

(c) When the jockey is moved from point A on the resistance wire to point C where $l = 30$ cm, find the increase in resistance of the circuit. (2 marks)